

AI Tools as Backbone

How **Ilm CLI**, **ComfyUI**, and **Langflow** support every entrepreneurship activity — without becoming the main event.

“Today we learn entrepreneurship. The AI helps us do it faster.”

Entrepreneurship First Rule — every session starts with the business goal, not the tool.

Generated from 2026 IFE Course Planning.xlsx · AI Tracks Comparison

How It Works

Your curriculum gives you 90 minutes of AI instruction per day inside a larger entrepreneurship program. The AI is a multiplier, not the subject. These three open-source tools replace the paid tools listed in the curriculum while keeping the learning objectives intact.

The Three Tools

TOOL	WHAT IT DOES	BEST FOR	REPLACES
CLI llm	Run prompts from the terminal. Pipe data in, get text out. Chain commands together.	Text generation, data analysis, roleplay, prompt chaining, batch exercises	ChatGPT Canvas, Perplexity Pro, Claude 4
UI ComfyUI	Node-based visual pipeline for AI image generation. Students see every step as a connected block.	Image creation, prototyping, visual bias comparison, pitch assets	Ideogram 2.0, Canva Magic Media, HeyGen
FLOW Langflow	Drag-and-drop builder for multi-agent workflows. Agents, tools, and LLMs as visual nodes.	Multi-agent research, chatbot building, pitch coaching, competitive analysis	Browse.ai, Typeform AI, Elicit

Order of introduction: Start with the CLI (Week 1). It's text-only and teaches the core skill: talking to a model. Add ComfyUI in Week 2 when you need images. Bring in Langflow in Week 3 when students are ready to orchestrate multiple agents. Each tool builds on the one before.

Week-by-Week Tool Map

Each row maps to the **Fun & Guided AI Track** column in your spreadsheet. The tool listed is the recommended replacement for the paid tool originally specified.

Week 1 — Foundations

DAY	ACTIVITY	TOOL	WHAT STUDENTS DO	ENTREPRENEURSHIP SKILL
1.2	The AI Whisperer	CLI	Students write prompts in the terminal to make the AI guess a secret word without using forbidden terms. They see how wording changes output.	Structured Prompting (T) · Probabilistic Logic (C)
1.3	Founder's Remix	UI	Build a ComfyUI pipeline that turns 3 random objects into an image of a social enterprise for a superhero. 10 wild ideas in 10 minutes.	Collaborative Design (C) · Multimodal Engineering (T)
1.4	The Domino Effect	CLI	<pre>cat problem_notes.txt llm -s "List 5 ridiculous + 5 serious consequences"</pre> Students pipe their research into the CLI to map second-order effects.	Root Cause Analysis (C) · Long-Context Management (T)

Week 2 — Discovery & Prototyping

DAY	ACTIVITY	TOOL	WHAT STUDENTS DO	ENTREPRENEURSHIP SKILL
2.1	The Grumpy User	CLI	<pre>llm -s "Roleplay as a skeptical 65-year-old..."</pre> Interview the persona, then switch to a friendly version. Learn how tone changes what people reveal.	Perspective Taking (C) · Role-play Prompting (T)
2.2	Treasure Hunt	FLOW	Build a Langflow agent that searches the web for real complaints about a topic. Students dig for "hidden gold" in what they find.	Qualitative Synthesis (C) · Autonomous Web Agents (T)
2.3	The Translator	CLI	Pipe messy team notes through chained llm calls: notes → haiku	

			→ rap → mission statement. One input, three outputs.	Information Architecture (C) · RAG-Based Retrieval (T)
2.5	Sketch-to-Magic	UI	Photo of a napkin sketch → IP-Adapter in ComfyUI → realistic app mockup. Students see their paper idea become a screen.	Modular Design (C) · Component Prompting (T)

Week 3 — Strategy & Systems

DAY	ACTIVITY	TOOL	WHAT STUDENTS DO	ENTREPRENEURSHIP SKILL
3.1	The Spy Agency	FLOW	Build a 3-agent Langflow graph: Researcher → Analyst → Writer. Each agent does one job. Students see the handoff between them.	SWOT 2.0 (C) · Multi-agent Research (T)
3.2	Robot Court	UI + CLI	Run the same prompt through different models side by side. ComfyUI for image models, CLI for text. Compare outputs and spot bias.	Meta-cognition (C) · Model Comparison (T)
3.3	The Money Tree	CLI	llm "List 5 weird ways people made \$1 from a similar idea" Pipe results into a Google Sheet for unit economics modeling.	Scalability Modeling (C) · Logic-Chain Formulae (T)
3.5	The 5-Minute Bot	FLOW	Build an emoji-only chatbot in Langflow using conditional nodes. Test it on other teams for "vibe checks."	Lean Methodology (C) · Chatbot Orchestration (T)

Week 4 — Markets, Money & Pitching

DAY	ACTIVITY	TOOL	WHAT STUDENTS DO	ENTREPRENEURSHIP SKILL
4.1	Time Traveler	CLI	llm -s "Write a 2040 front-page article about how our startup changed the world" Students craft the system prompt carefully to shape the narrative.	Trend Extrapolation (C) · Synthetic Data Simulation (T)
4.2	The Dragon's Hoard	CLI	llm "Calculate burn rate: \$X startup gold, spending \$Y/week like a	

			King vs \\$/week like an Inventor" Students run scenarios and compare.	Scenario Planning (C) · Spreadsheet Automation (T)
4.3	The Elevator Race	FLOW	Debate flow: Coach Agent vs Critic Agent vs Summarizer Agent. Students pitch, get feedback, revise, repeat. Fastest iteration wins.	Persuasive Communication (C) · Real-time Speech Analysis (T)

Week 5 — Finals

DAY	ACTIVITY	TOOL	WHAT STUDENTS DO	ENTREPRENEURSHIP SKILL
5.1–2	Hollywood Trailer	UI	Generate consistent pitch visuals through a ComfyUI pipeline. Export frames, add voiceover in CapCut (free).	Storytelling Rhetoric (C) · Video Synthesis (T)

Tool Setup Guides

Each guide is designed for school machines. No admin rights required for `pip install --user`. All three tools run on Mac, Windows, and Linux.

1. llm CLI

Installation

```
pip install llm or pip install --user llm
llm keys set openai (paste your API key)
llm keys set anthropic (paste your API key)
```

Quick Start for Students

```
llm "What is a stakeholder?" — basic chat
llm "Describe a park bench" --system "Use only 5 words" — system prompt
cat notes.txt | llm "Summarize this" — pipe data in
llm logs — review your history
```

Day 1 Script (5 minutes)

1. Open terminal. 2. Type `llm "hello"`. 3. Type `llm "hello" --system "reply in pirate speak"`. 4. Discuss: same input, different system prompt, completely different output. That's structured prompting.

No-Internet Fallback

Print prompt templates on cards. Students write prompts on paper. Facilitator reads the best one into the CLI. The class sees the output together.

2. ComfyUI

Installation

Download the portable version from github.com/comfyanonymous/ComfyUI/releases
Unzip and run. No Python setup needed for the portable build.

Quick Start for Students

Load the default workflow. It has one prompt node, one model node, one output node. Change the prompt, click "Queue Prompt," see the image appear. That's the whole mental model: prompt → model → output.

Key Workflows (Pre-built)

Sketch-to-Mockup: Load a photo → IP-Adapter node → prompt node → output. Turns napkin drawings into app screens.

Bias Comparison: Two model nodes side by side, same prompt, same seed. Compare what each model generates. Different results = different training data = bias you can see.

No-Internet Fallback

Print node graph screenshots on poster paper. Students trace the pipeline with markers. They move paper "tokens" through each node to simulate what the AI does.

3. Langflow

Installation

```
pip install langflow
```

```
python -m langflow run — opens a browser at localhost:7860
```

Quick Start for Students

Drag an "LLM" node onto the canvas. Connect it to a "Chat Input" node and a "Chat Output" node. Click play. You've built an AI. Now add a "Tool" node between input and output — that's an agent.

Key Flows (Export & Import)

Spy Agency (3 agents): Researcher Agent (searches the web) → Analyst Agent (finds weaknesses) → Writer Agent (writes a report). Each agent is one LLM node with a unique system prompt.

5-Minute Bot: Chat Input → Router Node (checks for emojis) → two paths: "valid emoji response" or "invalid, ask again."

Elevator Race: Student pitch → Coach Agent → Critic Agent → Summarizer → revised pitch.

No-Internet Fallback

Draw agent flowcharts on a whiteboard. Use sticky notes for each agent. Students physically move the sticky notes to show how a message travels through the system.

Quick Reference Card

WK	DAY	ACTIVITY	TOOL	SETUP TIME	ACTIVITY TIME	KEY PROMPT / FLOW
1	2	AI Whisperer	CLI	5 min	25 min	llm -s "Guess the word. Forbidden: X,Y,Z"
1	3	Founder's Remix	UI	10 min	30 min	Text-to-image workflow, 10 prompts
1	4	Domino Effect	CLI	5 min	25 min	cat notes llm -s "List consequences"
2	1	Grumpy User	CLI	5 min	25 min	llm -s "Roleplay as skeptical customer"
2	2	Treasure Hunt	FLOW	15 min	30 min	Agent with web search tool
2	3	The Translator	CLI	5 min	20 min	llm llm llm (chained)
2	5	Sketch-to-Magic	UI	15 min	35 min	IP-Adapter + prompt workflow
3	1	Spy Agency	FLOW	15 min	35 min	Researcher → Analyst → Writer
3	2	Robot Court	UI + CLI	10 min	40 min	Side-by-side model comparison
3	3	Money Tree	CLI	5 min	25 min	llm "5 weird ways to make \$1"
3	5	5-Minute Bot	FLOW	15 min	30 min	Emoji router with conditional nodes
4	1	Time Traveler	CLI	5 min	25 min	llm -s "Write 2040 article about..."
4	2	Dragon's Hoard	CLI	5 min	30 min	llm "Burn rate: X at Y vs Z"
4	3	Elevator Race	FLOW	15 min	35 min	Coach → Critic → Summarizer
5	1-2	Hollywood Trailer	UI	10 min	60 min	Consistent asset pipeline + CapCut

Entrepreneurship First Rule

Never open with the tool. Open with the problem.

Bad: "Today we're learning Langflow."

Good: "Today we're figuring out how big companies hide their weaknesses. Langflow will be our spy agency."

Bad: "Open the terminal and type this prompt."

Good: "Your customer won't tell you what's wrong. Let's build a persona that will."

The formula: "Today we learn *[entrepreneurship concept]*. The AI helps us *[action]*."

Offline / Low-Tech Fallbacks

SITUATION	WHAT TO DO
No internet	Print prompt templates on cards. Students write prompts by hand. Facilitator queues them up and runs them when connectivity returns. The thinking work is the same.
No laptops	Whiteboard + sticky notes. Draw agent flows manually. Students roleplay each agent. The person playing the "LLM" reads from a pre-printed response deck.
API keys blocked	Use <code>llm</code> with local models via <code>llm install llm-llama-cpp</code> . Runs entirely on the machine. Slower but works.
School firewall	ComfyUI and Langflow both run on <code>localhost</code> . No external requests needed once the models are downloaded.
Time runs short	Skip the setup. Use the CLI for everything. It takes 30 seconds to start and covers 60% of the activities.

IFE 2026 · AI Tools as Backbone · Generated from 2026 IFE Course Planning.xlsx
Tools: `llm` (datasette.io/llm) · ComfyUI (github.com/comfyanonymous/ComfyUI) · Langflow (langflow.org)